

ABC PUMPS - MODEL XYZ-200 TECHNICAL MANUAL

Section: 1. Technical Specifications

Equipment Tag: P-101 (Centrifugal Water Injection Pump)
Manufacturer: ABC Pumps Inc. | Model: XYZ-200 Heavy Duty
Design Flow Rate: 180 m³/h | Total Dynamic Head: 45 meters
Operating Speed: 2950 RPM | Impeller Diameter: 240 mm
Drive Motor: 75 kW Induction Motor (3-phase, 415V, 50Hz)
Installation Location: Plant A, Unit 2 - Fluid Processing Line

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Section: 2. Bearing Specifications & Clearances

Drive-End (DE) Bearing: Deep Groove Ball Bearing SKF 6312 C3

Non-Drive-End (NDE) Bearing: Cylindrical Roller Bearing NU 312

Recommended Radial Internal Clearance: 0.05 mm to 0.08 mm

Maximum Permissible Axial Float: 0.12 mm

Shaft Runout Tolerance: Less than 0.03 mm TIR at coupling hub

Torque Limits: Casing bolts 120 Nm, Bearing housing bolts 85 Nm

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Section: 3. Lubrication & Operating Limits

Approved Lubricant: ISO VG 46 Premium Synthetic Turbine Oil

Lubrication Sump Capacity: 2.4 Liters

Relubrication Interval: Every 4,000 operating hours or 6 months

Maximum Continuous Bearing Housing Temperature: 82°C (180°F)

Vibration Limit Baseline: ISO 10816-3 Class II pumps

Vibration Warning Threshold: 4.5 mm/s RMS (Velocity)

Vibration Trip / Shutdown Threshold: 9.0 mm/s RMS (Velocity)